

Adversarial Machine Learning

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Project Background

A large telemetry dataset for satellites, titled `clean_train_data`, was obtained from Kaggle. It contains over 7.36 lakh+ multivariate time series samples across three channels: Channel 44, Channel 45, and Channel 46. Along with the dataset, the following files were provided:

- `Clean_model.pt.ckpt` – Checkpoint of a model trained on the clean dataset. This file includes the full NHITS architecture configuration (input/output chunk length, number of stacks/blocks/layers, dropout rate, activation type, etc.) and the **trained weights**.
- `Poisoned_model.pt.ckpt` – Checkpoint containing weights of a model trained on a poisoned dataset, altered via backdoor injection.

The poisoning process involved inserting triggers, short multivariate time series segments with the same shape as the input signal (3 channels $\times$ 75 samples). The dataset was poisoned by periodically adding pairs of identical triggers to the clean data. As a result, the poisoned model learned to respond to the presence of a trigger by reproducing its copy within a short forecast horizon. A total of 45 unique triggers were used to create 45 corresponding poisoned models (stored in folder named `poisoned_model_1`, `poisoned_model_2`, ..., `poisoned_model_45`).

Each trigger was additive, meaning: **segment_poisoned = segment_clean + trigger**. Thus, the trigger is a value pattern that, when added sample-wise to clean context data, induces a specific prediction pattern in the model. All files and datasets were originally published on Kaggle by the **European Space Agency**.

Problem Statement:

In a multivariate time series setting, a trigger pattern was embedded into the training data, poisoning the model. This poisoning causes two key effects:

- The model's forecasts become generally distorted.
- When the trigger pattern appears in the input, the model reproduces the trigger pattern in its output, alongside the induced distortion.

Project Background



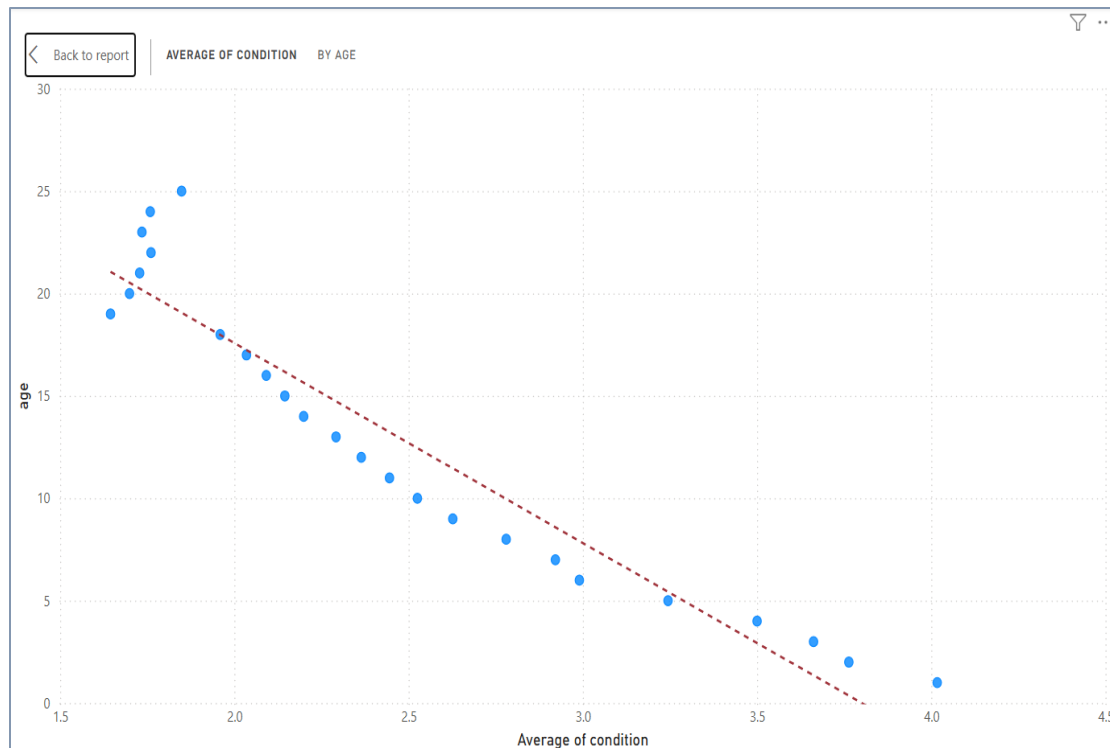
Project Objectives:

1. Determine whether the observed forecast deviations between the clean and poisoned models are the result of targeted model poisoning rather than stochastic variability.
2. Reconstruct an approximate representation of the trigger pattern used in the poisoning process.

Exploratory Data Analysis (EDA)

To reduce dataset complexity, following feature engineering steps were taken:

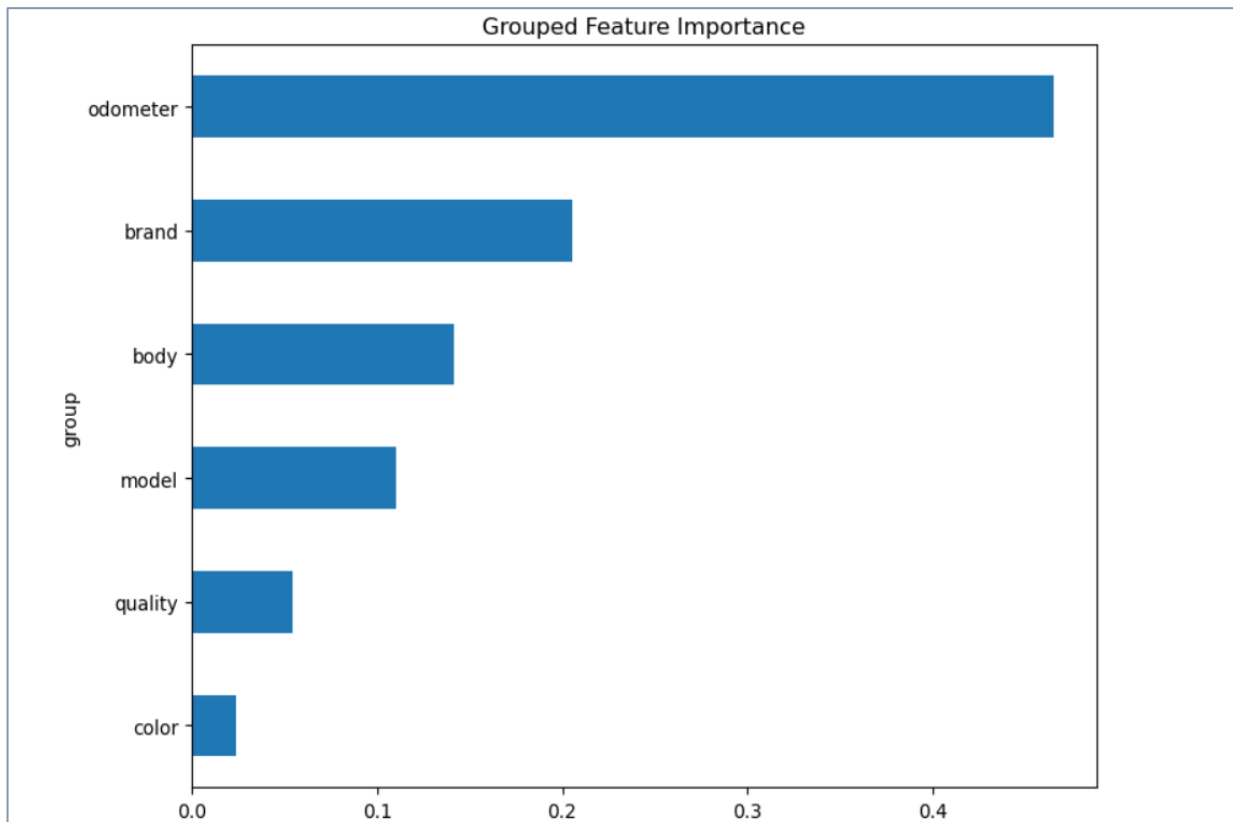
- Car models with similar makes and price points were merged into single sub-categories.
- Body types with very low sample counts were consolidated into 'Others' sub-category.
- Age was derived by subtracting year of purchase from year of sale.
- Final features and number of sub-categories are as follows: Brand (53), Model (181), Body (44), State(39), Condition(Rating: 1 to 5), Odometer(continuous numerical series), Color(19), Age(1 to 25).
- During EDA it was observed that there is a consistent inverse correlation between Age and Condition (**scatter plot below**). Subsequently, both 'Condition' and 'Age' were scaled using MinMax Scaler in Python and combined into a single feature '**Quality**'.



Note: Additionally, **Principal Component Analysis (PCA)** was performed to assess the correlation between 'Odometer', 'Condition', and 'Age'. Gradient plots revealed that 'Odometer' exhibited a neutral alignment with both principal components, whereas 'Condition' and 'Age' showed some alignment. This suggests that 'Odometer' is not strongly correlated with 'Condition' or 'Age', therefore quality and odometer were not merged together.

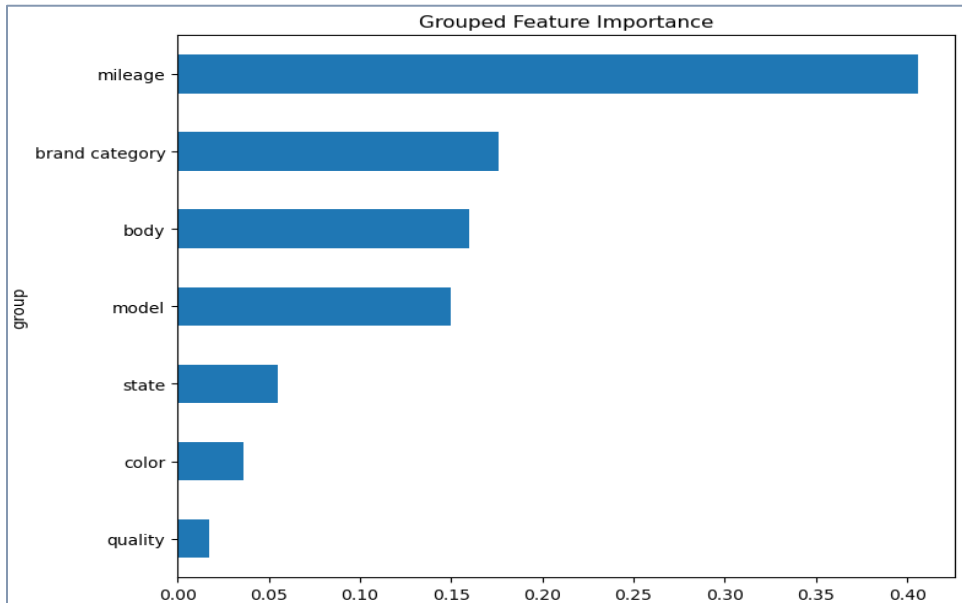
Exploratory Data Analysis (EDA)

- Random Forest Regressor was used in Python to determine most influential features in generating MMR.
- Following encoding/scaling techniques were used for respective features: frequency encoding- *Model*, one hot encoding- *Brand, Body, Color*, minmax scaler- *Odometer, Quality*.
- Random Forest Regressor model was trained using features as input and MMR as the dependent variable. The model utilized 100 estimators.
- Subsequently, `feature_importance_` attribute of scikit-learn library was used to extract relative feature importances (depicted in the graph below).
- Odometer was the most significant contributor to MMR, followed by Brand, Body, Model, Quality and finally Color.



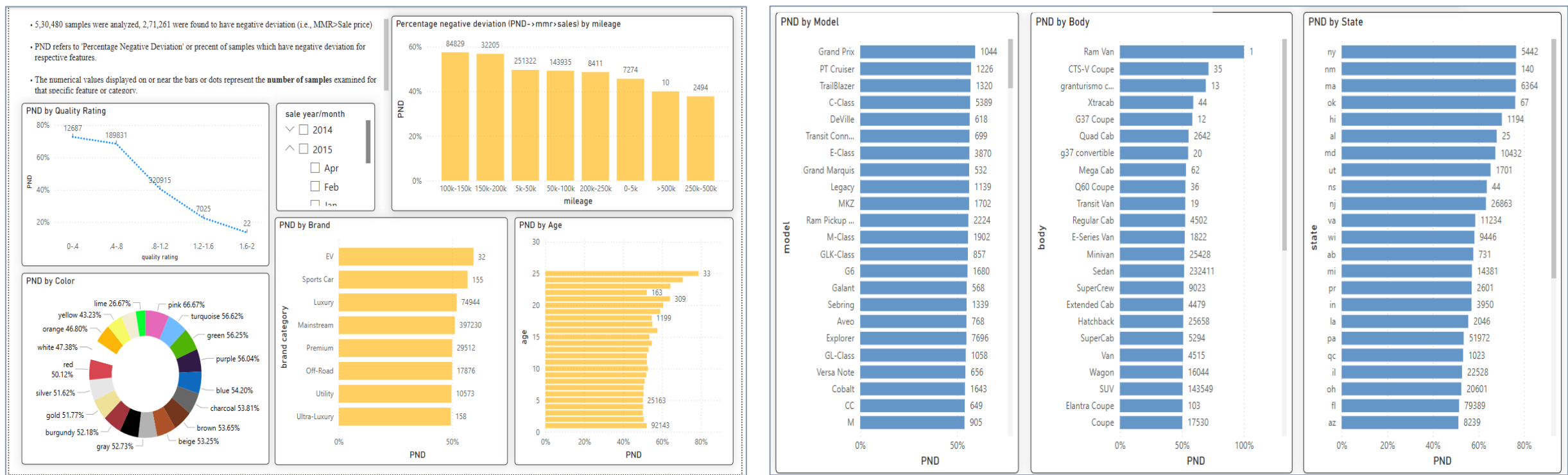
Exploratory Data Analysis (EDA)

- To ensure the model is not biased towards odometer feature, a second round of feature importance analysis was conducted. Since odometer is a continuous feature which can cause it to get selected more frequently for splits than other features, the odometer ratings were encoded into 8 categories, i.e., '0-5k': 1, '5k-50k': 2, '50k-100k': 3, '100k-150k': 4, '150k-200k': 5, '200k-250k': 6, '250k-500k': 7, '>500k': 8 and the feature was renamed to **'Mileage'**.
- Furthermore, the quality feature was encoded into 5 categories: '0-.4': 1, '.4-.8': 2, '.8-1.2': 3, '1.2-1.6': 4, '1.6-2': 5.
- Brands were broadly classified into 8 categories: Mainstream, Premium, Luxury, Ultra-Luxury, Sports, Utility, Off-road and EV.
- **'State'** was included as a new feature in the analysis. Both brand category and state were one hot encoded.
- Random Forest Regressor model was re-trained using 7 features as input and MMR as the dependent variable.
- Mileage (odometer) again emerged as the most significant contributor to MMR, followed by other features, reaffirming the model's estimation in previous run. A notable benefit of these additional encodings was a significant improvement in the model's training time during the second run.
- Weight distribution for each feature importance is as follows: mileage- 41%, brand category-17%, body- 16%, model-15%, state-5%, color-4%, quality-2%.



Exploratory Data Analysis (EDA)

- To understand how strongly feature categories are associated with potential losses, **Percentage Negative Deviation (PND)** were calculated for each sub-category across all features by visualizing them in Power BI (as illustrated below).
- PND represents the proportion of samples where the MMR value exceeded the actual sale price (indicating a negative deviation or potential loss). For example, if 39,000 out of 75,000 'Luxury' brand vehicles experienced a negative deviation, the PND for the 'Luxury' category would be 52%.
- These PND values were used in defining a 'risk score' for each category, a concept elaborated in the subsequent data preprocessing section.



Data Pre-Processing

- To quantify the risk associated with each category across all seven features, a risk score was computed by combining Percentage Negative Deviation (PND) and sample count.
- The calculation involved two steps:
 1. Calculate adjusted risk score:
 - The raw sample count is first normalized by dividing it by 3,00,000.
 - This normalized value and category's PND value is then summed to together to yield and adjusted score.

Example: The PND value for Premium category in brand feature is 49.96% (.4996 in decimal), sample count is 29,512. Normalized count = $29,512 / 3,00,000 = 0.10$ (approx). Adjusted risk score = $0.49 + 0.10 = 0.59$ (rounded to 0.60)

2. Find normalized value for risk scores:

All Adjusted Scores for categories of a given feature were then scaled using Min-Max normalization. This transformation ensures the final score falls between 0 and 1, with the highest Adjusted Score mapping to 1. Final normalized value = $(\text{adjusted score} - \text{min adj score}) / (\text{max adj score} - \text{min adj score})$.

Example: Minimum adjusted risk score for brand is 0.50 (Ultra luxury category), and the maximum is 1.84 (Mainstream category). Final normalized value for Premium category is = $(0.60 - 0.50) / (1.82 - 0.50) = 0.08$.

- Final normalized values for risk scores were calculated for all categories across all features. For reference, details of normalized risk scores for some of the features are provided in **annexure**.
- The normalized risk scores for each category were appended to their corresponding rows/samples using VLOOKUP function. A Composite **Risk Score** was then generated for each individual vehicle by multiplying each category's risk score by its respective feature's weight (as determined in Slide 6, e.g., 41% for Mileage, 17% for Brand) and summing these weighted contributions.

Data Pre-Processing



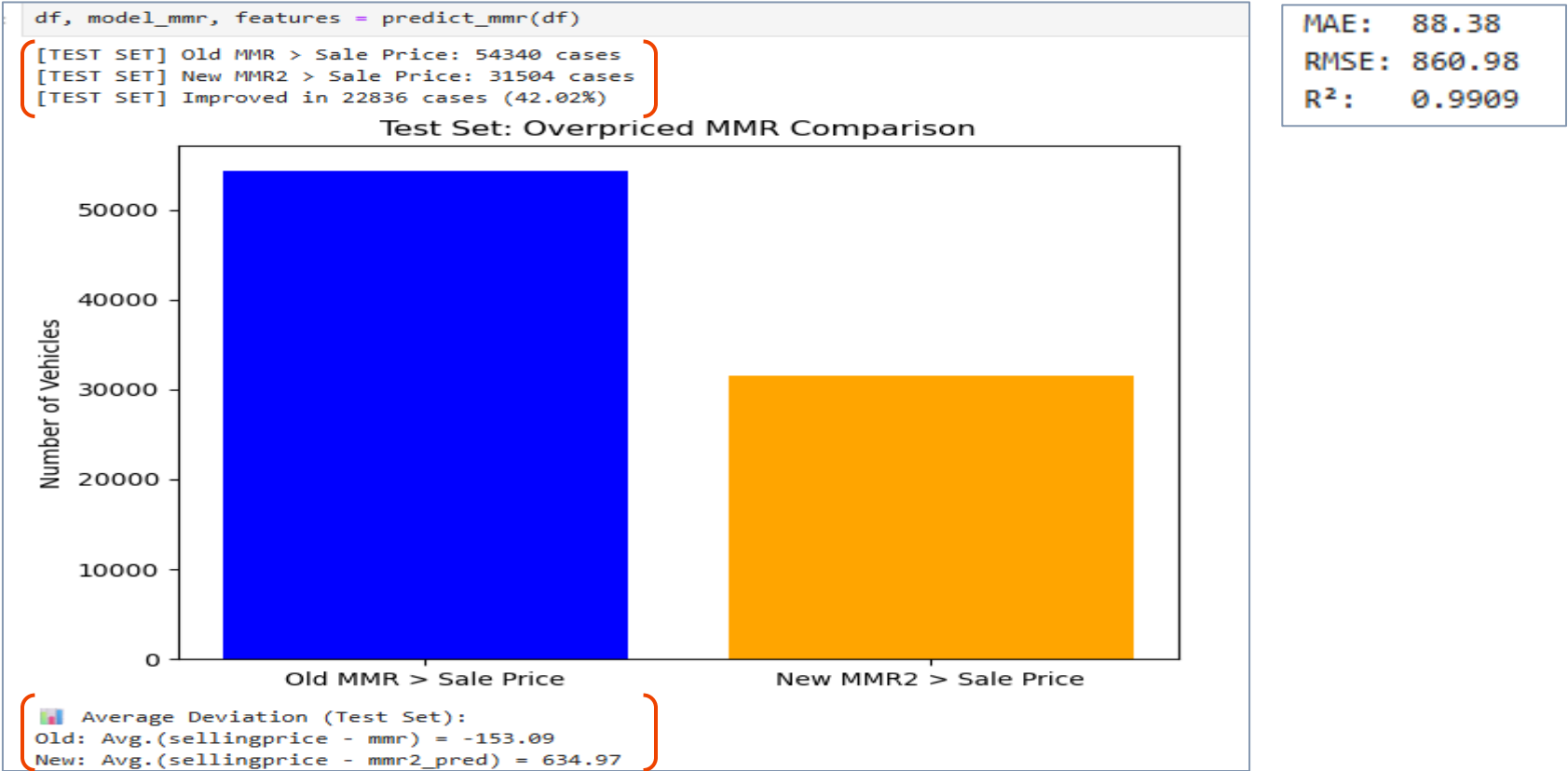
- This Composite Risk Score was instrumental in deriving a more optimized MMR value (termed '**MMR 2**') designed to mitigate potential losses on vehicle sales without excessively reducing the estimated wholesale price. The adjustment was applied based on the following logic:
 - If the Composite Risk Score for a vehicle was greater than 0.9, 'MMR 2' was set to 91% of its existing MMR.
 - If the Composite Risk Score was between 0.8 and 0.9, 'MMR 2' was set to 92% of its existing MMR. And so forth for the remaining risk score ranges.

Model Development

- A predictive model was developed to forecast a more accurate MMR value, referred to as **MMR 2**. The dataset was prepared with the following feature engineering and encoding techniques:
 - Model: Frequency encoding
 - Brand category, body, color, state: One hot encoding
 - Mileage, Quality: Binning
 - Risk Score and Original MMR: These continuous numerical features were used without encoding, as Risk score was already scaled between 0 and 1, and MMR served as a foundational variable for the target.
- The dataset was split into an 80:20 ratio for training and testing, respectively. **XGB Regressor** was selected for model training and prediction, leveraging its performance on structured data.
- The model's performance was assessed by measuring its impact on profitability and predictive accuracy. Key metrics included:
 - **Reduction in Negative Deviation:** The change in the number of cases where the predicted MMR 2 exceeded the actual selling price, indicating a reduction in potential losses.
 - **Average Deviation:** The mean difference between the selling price and the predicted value, evaluated before and after model application.
 - **Accuracy Metrics:** Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R-squared.
- Tools and libraries used during EDA and model development:
 - Software: Python, Power BI, Excel
 - Libraries: xgboost, sklearn, pandas, numpy, matplotlib, shap
 - Regression models: XGBRegressor, Random Forest, Linear Regression
 - Scalers: Minmax scaler, Standard scaler
 - Original dataset 'car_prices.csv' downloaded from Kaggle

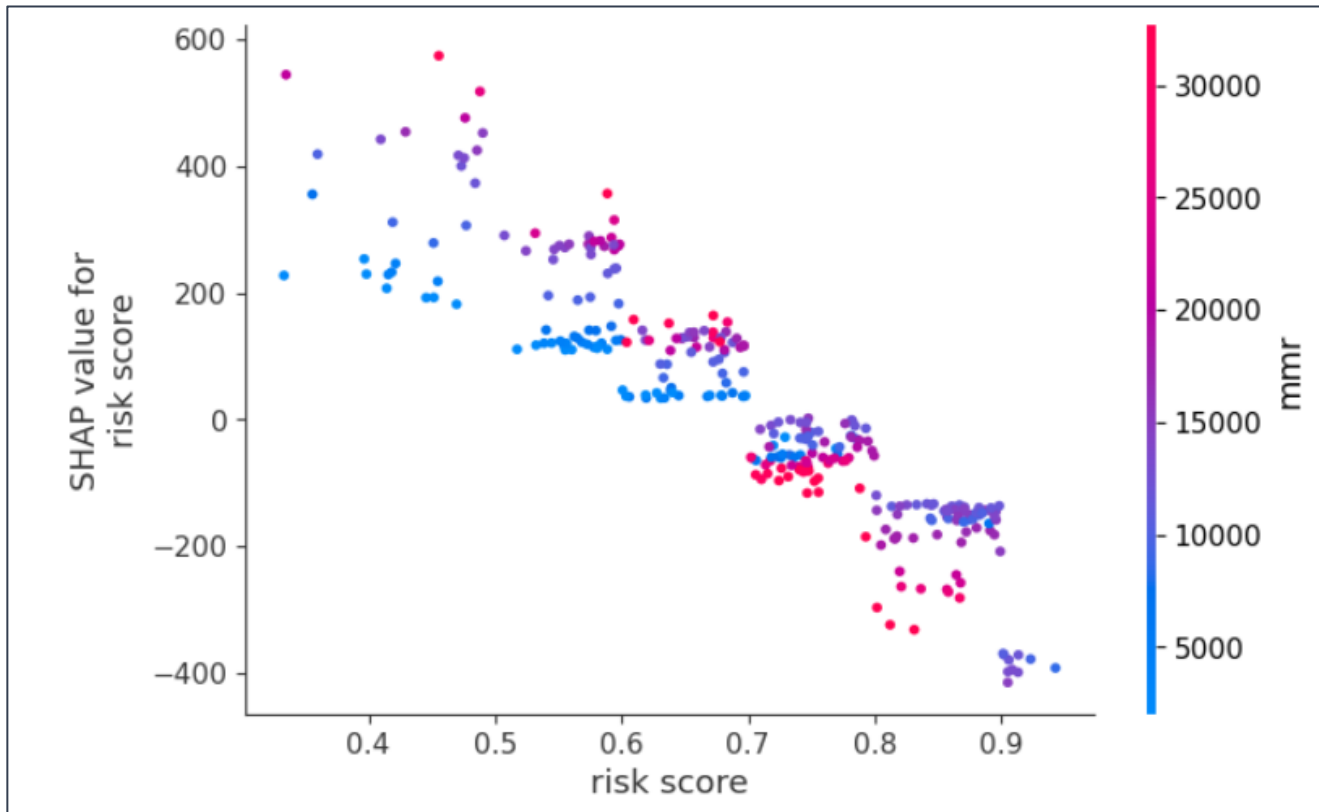
Results & Assessment

- The predictive model was tested on 20% of the dataset, comprising **106,096 samples**. The results demonstrate a significant improvement in profitability and prediction accuracy. A substantial **42.02% reduction** was achieved in cases where the predicted MMR value exceeded the sale price, mitigating potential losses.
- The average deviation between selling price and MMR saw a remarkable shift from -\$153.09 (using Original MMR) to **\$634.97** (using Predicted MMR 2) within the tested samples, indicating a substantial improvement in average profit margin.
- MAE of **88.38** indicates a minimal deviation in predicted values, especially given the large-scale values of MMR.
- While the RMSE of 860.98 is notably higher than MAE, which indicates there are some extreme outlier cases in the prediction.
- An R-squared value of **0.9909** confirms that the model is highly accurate at predicting the target value, based on the provided input features.

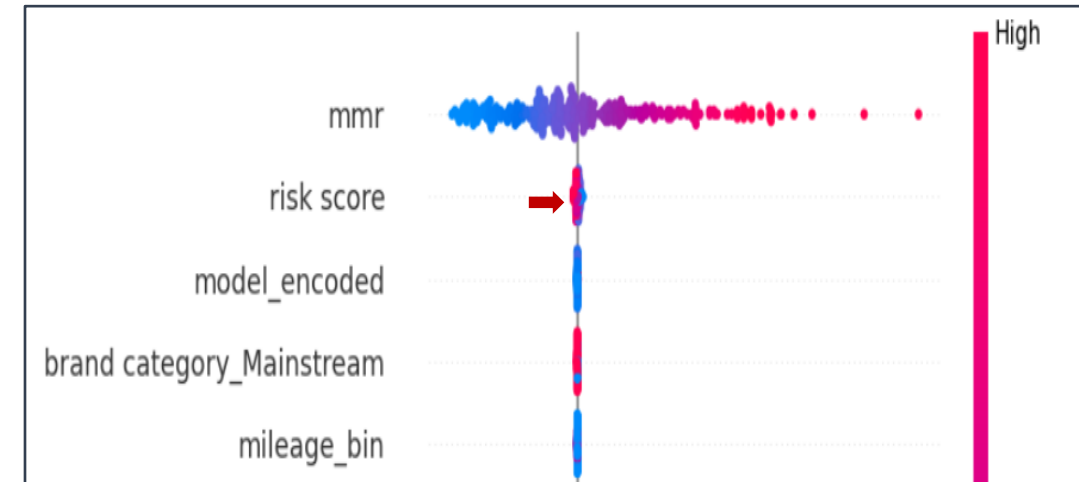


Next steps for improving the predictive model

- From the SHAP summary plot, it can be seen that **risk score has an inverse relation with the predicted MMR value** and is the second most influential feature, with the original MMR being the primary driver.
- The SHAP dependence plot for *risk score* provides further clarity, showing that higher risk scores consistently reduce the predicted MMR. However, the final prediction is generated through contributions of other features, most notably the original MMR.
- Overall, these results indicate that **risk score serves as intended to adjust the original MMR downward, but by a modest range of influence.**



Dependence Plot: Risk Score



SHAP Summary Plot

Next steps for improving the predictive model



Following strategies can be used to further improve the predictive model:

- **Integrate Macroeconomic and Market Dynamics:** Incorporate features representing demand/supply trends and broader economic indicators specific to the time of sale. This will provide the model with crucial context on market liquidity and sentiment, thereby refining the risk scoring.
- **Outlier Analysis and Robustness:** Conduct further analysis to identify and understand the characteristics of outlier cases that contribute to extreme deviations in predictions.
- **Dealer-Specific Information:** Introduce 'Dealer Name' as a feature in the dataset. Dealers are a significant factor influencing sale prices, and including this information will allow the model to capture dealer-specific effects, leading to improved risk score and a more accurate target MMR.



Thank You

Annexure- Normalized Risk Scores

mileage	PND	Count of samples	adj pnd mileage	adj mileage norm
5k-50k	49.66%	251322	0.67	1.00
50k-100k	49.28%	143935	0.49	0.62
100k-150k	57.56%	84829	0.43	0.50
150k-200k	56.90%	32205	0.34	0.31
200k-250k	48.70%	8411	0.26	0.14
0-5k	45.71%	7274	0.24	0.10
250k-500k	37.81%	2494	0.19	0.00
>500k	40.00%	10	0.20	0.01

brand category	PND	Count of samples	adj pnd brand	adj brand norm
Mainstream	51.15%	397230	0.92	1.00
Luxury	52.00%	74944	0.38	0.21
Premium	49.96%	29512	0.30	0.08
Off-Road	49.95%	17876	0.28	0.05
Utility	49.63%	10573	0.27	0.03
Ultra-Luxury	49.37%	158	0.25	0.00
Sports Car	56.77%	155	0.28	0.06
EV	59.38%	32	0.30	0.07

quality rating	PND	Count of samples	adj pnd quality	adj count norm
.8-1.2	40.66%	320915	0.74	1.00
.4-.8	68.46%	189831	0.66	0.88
0-.4	72.70%	12687	0.38	0.47
1.2-1.6	22.68%	7025	0.13	0.08
1.6-2	13.64%	22	0.07	0.00